THE RING OF RATIONAL FUNCTIONS OF A SCHEME WITH FINITELY MANY IRREDUCIBLE COMPONENTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Rational functions as a colimit of global sections of open dense subschemes. Schemes, Example \ref{schemes-example-global-sections}, cited at proof lines 12734-12736. Identifies the ring of rational functions with the colimit of the rings of global sections.
- The local ring at the generic point of a reduced scheme is its residue field. Algebra, Lemma \ref{algebra-lemma-minimal-prime-reduced-ring}, cited at proof lines 12737-12738. Gives the reduced and integral final statements.

Source reference. Original lemma 01RV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2316–2316. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01JH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2014–2014. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00EU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 548–548. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let X be a scheme with finitely many irreducible components $X_1, \ldots, X_n$. If $\eta_i \in X_i$ is the generic point, then*

$$R(X) = \mathcal{O}_{X,\eta_1} \times \ldots \times \mathcal{O}_{X,\eta_n}$$

If X is reduced this is equal to $\prod \kappa(\eta_i)$. If X is integral then $R(X) = \mathcal{O}_{X,\eta} = \kappa(\eta)$ is a field.

Proof. Let $U \subset X$ be an open dense subset. Then $U_i = (U \cap X_i) \setminus (\bigcup_{j \neq i} X_j)$ is nonempty open as it contained η_i, contained in X_i, and $\bigcup U_i \subset U \subset X$ is dense. Thus the identification in the lemma comes from the string of equalities

$$\begin{aligned} R(X) &= \operatorname{colim}_{U \subset X \text{ open dense}} \operatorname{Mor}(U, \mathbf{A}^1_{\mathbf{Z}}) \\ &= \operatorname{colim}_{U \subset X \text{ open dense}} \mathcal{O}_X(U) \\ &= \operatorname{colim}_{\eta_i \in U_i \subset X \text{ open}} \prod \mathcal{O}_X(U_i) \\ &= \prod \operatorname{colim}_{\eta_i \in U_i \subset X \text{ open}} \mathcal{O}_X(U_i) \\ &= \prod \mathcal{O}_{X,\eta_i} \end{aligned}$$

where the second equality is Schemes, Example 01JH. The final statement follows from Algebra, Lemma 00EU. □